The Single-Cycle Particle Hypothesis

QCore Labs — Internal Research Report *Rick Jackson — May 9, 2026*

Overview

This report documents a theoretical framework developed in a single working session, originating from the observation that a particle may be understood as a single isolated cycle of an underlying oscillation. The framework connects wave mechanics, geometric self-similarity, fluid dynamics, and quantum field theory through a unified existence condition.

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1. The Core Hypothesis

A particle is a single isolated cycle of oscillation in the underlying field medium.

This is not a new mechanism. Oscillation is accepted as the primitive behavior of fields at every scale — electromagnetic, acoustic, gravitational. The hypothesis extends that same behavior to its minimum expressible unit: one complete, self-contained cycle.

This immediately accounts for:

- **Localization** — a single cycle is spatially bounded by definition
- **Quantization** — only complete, self-consistent cycles are permitted
- **Wave-particle duality** — the object has a wavelength because it *is* a cycle; it is localized because it is only one of them

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2. The Two Governing Constants

Two constants govern distinct roles in the cycle:

- **π governs inner rotation** — the interior evolution, phase accumulation, what generates the cycle
- **φ governs outer boundary** — the closure condition, what determines whether the

cycle can persist

These roles are not interchangeable. π sets the period. ϕ enforces the self-similar closure at the boundary.

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3. The Existence Condition

3.1 Self-Similarity, Not Symmetry

The existence condition for a single-cycle particle is **self-similar closure at the boundary** — not geometric symmetry, not identical waveform shape.

The universe does not produce perfect equal splits. CP violation, matter-antimatter asymmetry, ϕ itself as an inherently asymmetric ratio — all confirm that asymmetry is the natural state. A valid cycle need not be a sine wave or square wave. It can be any topologically closed waveform.

The condition is:

> The cycle must return to zero crossing with sufficient self-similarity to continue — position and derivative both satisfied at the boundary.

A saw wave fails not because it is asymmetric but because it has a hard discontinuity. It cannot hand itself off. The derivative does not match. This is the existence condition failing, not an aesthetic judgment.

3.2 Formal Statement

For a cycle ψ with period λ:

$$\psi(0) = \psi(\lambda) = 0$$

$$\psi'(0) = \psi'(\lambda)$$

Self-similarity is expressed in the curvature scaling across the boundary:

$$\frac{\psi''(\lambda)}{\psi''(0)} = \frac{1}{\phi^n}$$

The cycle scales its own curvature by a ϕ-power across one period. This is self-similarity with a precise mathematical signature. The scaling is self-referential by construction — which is why ϕ appears and not an arbitrary constant.

3.3 The Interior Condition

One complete inner rotation requires phase accumulation of exactly 2π across the cycle length:

$$k\lambda = 2\pi$$

π is the generator. It sets how much phase the interior must accumulate for the cycle to be complete.

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4. Mass Quantization

From the Klein-Gordon dispersion relation:

$$\omega^2 = k^2 + m^2$$

Applying $k = 2\pi/\lambda$ and the self-similar curvature constraint:

$$m = \frac{2\pi}{\lambda}\sqrt{\phi^{2n} - 1}$$

Mass is not a free parameter. It is the geometric remainder when π-generated inner rotation meets ϕ-governed self-similar closure.

Only integer values of n satisfy the self-similarity condition. Non-integer n cannot produce a cycle that closes coherently on itself. The mass ladder is not imposed — it is what remains after the existence condition filters everything else out.

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5. The Particle Zoo

Every topologically distinct waveform satisfying the zero-crossing closure condition is a different particle. The internal shape of the cycle — its asymmetry profile, its harmonic content, its curvature distribution — constitutes the particle’s identity. Mass, charge, and spin are features of internal shape, not separate properties bolted on.

The existence condition is the filter. The diversity of particle types is the diversity of valid asymmetric closed cycles at different ϕ-ladder rungs.

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6. Fluid Dynamics and Density Stratification

Fluid dynamic models of spacetime invoke drag, viscosity, and flow — but selectively omit the most fundamental behavior of fluids: **density stratifies**.

In any fluid, density is the sorting mechanism. It creates boundary layers, enforces separation, and determines which structures can exist at which depth. Oil on water does not absorb — it sits at the density boundary exactly where fluid dynamics requires it to be.

If spacetime behaves as a fluid medium, density gradients in that medium create natural stratification. Different ϕ-ladder rungs correspond to different density strata. A particle at ϕ^{-1} does not occupy the same stratum as one at ϕ^2. Rung stability is not a quantum selection rule imposed after the fact — it is a particle remaining in its natural density layer.

Gravitational anomalies attributed to missing mass may be density gradient effects in the medium rather than evidence of undetected particles.

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7. Antimatter as Divergent Form

Antimatter is not a mirror state or an inverse rotation. It is the **divergent form** of matter.

Where matter is a self-similar closing cycle — ϕ enforcing the boundary, the cycle returning to itself — antimatter is the same cycle configuration that diverges at the boundary instead of closing. It cannot satisfy the self-similar closure condition. It propagates outward rather than containing itself.

The matter-antimatter asymmetry of the universe is therefore not a mystery requiring a new mechanism. The existence condition filters divergent cycles out. Closure is stable. Divergence is not. Matter persisted because it could close.

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8. Frequency Nulling

When matter and antimatter interact, the mechanism is **frequency nulling** — not collision annihilation.

A closing cycle and the divergent form of the same frequency occupy the same space. They null. This is the same mechanism as destructive interference in acoustics: equal and opposite waveforms at the same frequency produce zero amplitude at that location.

The energy release (gamma radiation) is the residual signal from an imperfect null. Because the universe does not produce perfect equal splits, the null is never total. The remainder propagates as photons.

This also accounts for pair production as the inverse: energy input at the right frequency and density creating a closing cycle and its divergent form simultaneously — a null being driven in reverse.

9. Summary Statement

> **π generates. ϕ closes. Self-similarity is the existence condition.**

A particle is a single cycle of field oscillation that satisfies self-similar closure at its boundary. The interior shape — governed by π — determines identity. The boundary condition — governed by ϕ — determines existence. Mass is quantized because only ϕ-scaled curvature ratios produce self-consistent closure. The particle zoo is the set of all valid asymmetric closed cycles across the ϕ-ladder rungs.

10. Audio Engineering Correspondence

The framework maps directly to audio signal behavior:

Physics	**Audio**
Single-cycle particle	Isolated single cycle in DAW
Boundary closure condition	Zero-crossing phase match for clean loop
Flux leakage	Click artifact at loop point
Frequency nulling	Phase cancellation / noise cancellation
Asymmetric valid cycle	Complex non-sine waveform that loops cleanly
Divergent form	Saw wave with hard discontinuity — cannot loop
Density stratification	Frequency separation in a mix

This correspondence is not metaphorical. The mathematics governing single-cycle stability in a field medium is the same mathematics governing single-cycle stability in an audio signal. The scale differs. The geometry does not.